

PYTHON I 12 - 16

Lesson 16. Link Converter

Lesson description

Students will create an application that will shorten the link from any website that they will gather. the application that students will create is still using the library tkinter.

Tools:

1. PyCharm

Learning Objectives

Students will:

1. Design the application
2. Implement their knowledge about placement
3. Create a function for each button
4. create a function for entry fields
5. import library request to convert the link
6. import library pyperclip to copy text from application
7. Students will learn about error handler.

Timing	Lesson description	Additional materials	Comments
5 min	Preparing • Guide students to prepare the project directory. • Tell students to open Pycharm • Make sure Internet Connection is Good • Review the lesson		

5 min	Comprehension check • We ask students to write the things they have learned during the last lesson on the board. • We ask students the syntax that we use in the last lesson. For example, we can ask students if they still remember to creating a button, label and also entry fields		
10 min	Error Handler Explain and Code • The teacher shows the result of the program. and tell the students today we will use the error handler to handle the error of the application. • try is used to test your block of code for errors. exception is used to show the error of your code. • Teachers tell the students that we can control the error and put the condition that we want using this handler. • let the students try by themself and explore it. • ask the students how many possibilities that we can use using the error handler.	1. Error Handler.py	
10 min	Shorten the Link Explain and Code • Teacher tells the students that we need to import library requests. this library will request the HTTP from the website. • Teachers tell the students that using this library we can request the HTTP to shorten the link. • Teachers tell the students we will use the API from the website https://cutt.ly/.	2. Link Converter.py	

	- Let the students login into this website and tell them to get the API from this website. - tell the students they can find the API from the General - API - API Key. - Teacher tells the students to copy the API key. - and copy the link URL Shortener from the API Documentation to do the request shorten the link.		
10 min	Variable Explain and Code API_KEY is the identifier that will be used to access your account to manipulate the process. BASE_URL is the API URL that will be used to do the request from the website. At this moment the request is to shorten the link. payload is to gather all the requests from the API_KEY to shorten and modify the name of the link that already shorten. DATA is to read the request body.		
45 min	Shorten Link Tkinter Explain and Code - Teacher shows the project and ask the students if they still remember to create the widget of the project shown. - Let the students design the application. - After that, tell the students that we will use the function that was already created before but we need some adjustment because the Tkinter widgets need to catch the function.	3. Link Converter Tkinter.py	

	• Because we use the Tkinter library to design the Application. we should show the error using the Message Box. • Ask the student if they still remember about the message box. • let the students put the message box to handle the error.		
5	Last Overview Summing Up • We sum up by asking questions making sure the students remember the most important things. • Hands up who understands what arrays are and how we can use them in our program? • We thank students for doing a great job overall.		